

Ordinary Differential Equations

A First-Principles Cheat Sheet for Second-Order Linear Systems & Mechanical Vibrations

Color key: all live solution steps in the **Worked Examples** are typeset in purple; explanatory text stays black. Every system here is governed by one master equation — Newton's second law applied to a mass on a spring:

$$m\ddot{x} + c\dot{x} + kx = F(t),$$

where $x(t)$ is displacement from equilibrium. Each section below is a different *shape* this equation can take: turning off $F(t)$ gives free vibration; turning off c gives undamped motion; matching the driving frequency to the natural frequency gives resonance.

Notation at a glance

m, c, k	mass, damping coefficient, spring constant (all > 0 ; c may be 0).
$\omega_0 = \sqrt{k/m}$	natural (undamped) angular frequency.
$\beta = \frac{c}{2m}$	damping factor (decay rate of the envelope).
$\omega_1 = \sqrt{\omega_0^2 - \beta^2}$	quasi-frequency of damped oscillation (underdamped only).
$c_{\text{cr}} = 2\sqrt{mk} = 2m\omega_0$	critical damping coefficient (boundary between regimes).
ω, F_0	driving angular frequency and forcing amplitude in $F(t) = F_0 \cos \omega t$.

1. Simple Harmonic Motion (SHM) & Undamped Free Vibrations

The Core Logic

Hang a mass m on a spring with stiffness k . **Hooke's Law** says the spring pushes back in proportion to stretch, $F_{\text{spring}} = -kx$. With no damping ($c = 0$) and no driving ($F = 0$), Newton's second law $m\ddot{x} = \sum F$ becomes

$$m\ddot{x} = -kx \implies \ddot{x} + \omega_0^2 x = 0, \quad \omega_0 = \sqrt{\frac{k}{m}}.$$

The characteristic equation $r^2 + \omega_0^2 = 0$ has purely imaginary roots $r = \pm i\omega_0$ — that is *why* the motion is sinusoidal. The general solution is

$$x(t) = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t = A \cos(\omega_0 t - \phi),$$

with **amplitude** $A = \sqrt{c_1^2 + c_2^2}$ and **phase** $\tan \phi = c_2/c_1$ (choose the quadrant so that $A > 0$). Motion is periodic forever — energy is conserved.

Vertical-spring note. Gravity merely shifts the equilibrium: at rest $mg = kL$ (static stretch L). Measuring x from the *new* equilibrium cancels the mg term, so the ODE is unchanged. This is how a physical spring problem becomes $\ddot{x} + \omega_0^2 x = 0$.

All Possible Cases / Variations

- **Horizontal vs. vertical spring.** Same ODE about equilibrium — vertical springs only require the extra step $k = mg/L$.
- **Small-angle pendulum.** $\ddot{\theta} + \frac{g}{L}\theta = 0$ has the identical form with $\omega_0 = \sqrt{g/L}$.
- **Two solution forms.** Rectangular (c_1, c_2) for applying initial conditions; amplitude-phase (A, ϕ) for reading off physical behavior. They are interchangeable.
- **Frequency vocabulary.** Angular frequency ω_0 (rad/s), ordinary frequency $f = \omega_0/2\pi$ (Hz), period $T = 2\pi/\omega_0$.
- **Initial Value Problem (IVP).** $x(0) = x_0$ and $\dot{x}(0) = v_0$ pin down $c_1 = x_0$ and $c_2 = v_0/\omega_0$.

Step-by-Step Methodology

1. Model the forces: write $m\ddot{x} + kx = 0$ from Newton's law and Hooke's law.
2. Compute $\omega_0 = \sqrt{k/m}$; for a vertical spring extract k from $mg = kL$ first.
3. Write $x = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t$.
4. Apply $x(0) = x_0 (\Rightarrow c_1)$ and $\dot{x}(0) = v_0 (\Rightarrow c_2 = v_0/\omega_0)$.
5. (Optional) Convert to $A \cos(\omega_0 t - \phi)$ for amplitude and phase.
6. Report period $T = 2\pi/\omega_0$, frequency, and amplitude.

Worked Example A — From Physical Data to ω_0

A mass $m = 2$ kg stretches a spring $L = 0.5$ m to its equilibrium position (use $g = 10$ m/s²). Find the natural frequency and period.

Step 1 — Spring constant from static stretch ($mg = kL$):

$$k = \frac{mg}{L} = \frac{(2)(10)}{0.5} = 40 \text{ N/m.}$$

Step 2 — Natural frequency :

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{40}{2}} = \sqrt{20} = 2\sqrt{5} \approx 4.47 \text{ rad/s.}$$

Step 3 — Period :

$$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{2\sqrt{5}} = \frac{\pi}{\sqrt{5}} \approx 1.40 \text{ s.}$$

The mass oscillates forever at ≈ 4.47 rad/s.

Worked Example B — Undamped IVP & Amplitude–Phase

Solve $\ddot{x} + 9x = 0$ with $x(0) = 4$, $\dot{x}(0) = -6$.

Step 1 — Natural frequency & general solution ($\omega_0 = 3$):

$$x(t) = c_1 \cos 3t + c_2 \sin 3t.$$

Step 2 — Apply $x(0) = 4$:

$$x(0) = c_1 = 4.$$

Step 3 — Differentiate and apply $\dot{x}(0) = -6$:

$$\dot{x}(t) = -3c_1 \sin 3t + 3c_2 \cos 3t, \quad \dot{x}(0) = 3c_2 = -6 \Rightarrow c_2 = -2.$$

Step 4 — Particular solution :

$$x(t) = 4 \cos 3t - 2 \sin 3t.$$

Step 5 — Convert to amplitude–phase :

$$A = \sqrt{4^2 + (-2)^2} = 2\sqrt{5}, \quad \tan \phi = \frac{c_2}{c_1} = -\frac{1}{2} \Rightarrow \phi \approx -0.4636 \text{ rad.}$$

$$x(t) = 2\sqrt{5} \cos(3t + 0.4636), \quad T = \frac{2\pi}{3}.$$

2. Damped Free Vibrations

The Core Logic

A dashpot adds a resistive force proportional to velocity, $F_{\text{damp}} = -c\dot{x}$. Newton's law now gives the **homogeneous** equation

$$m\ddot{x} + c\dot{x} + kx = 0 \implies \ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0, \quad \beta = \frac{c}{2m}.$$

The characteristic equation $mr^2 + cr + k = 0$ has roots

$$r = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}.$$

Everything hinges on the **discriminant** $c^2 - 4mk$ —equivalently on c versus the critical value $c_{\text{cr}} = 2\sqrt{mk}$.

All Possible Cases / Variations

- **Overdamped** ($c^2 > 4mk$): two distinct real *negative* roots. No oscillation—a slow crawl back to equilibrium, crossing zero at most once.
- **Critically damped** ($c^2 = 4mk$): one repeated root $r = -\beta = -\omega_0$. Fastest return to rest without overshoot—the engineering ideal (door closers, shock absorbers).
- **Underdamped** ($c^2 < 4mk$): complex roots $-\beta \pm i\omega_1$. Oscillation trapped inside a decaying envelope $\pm A e^{-\beta t}$.
- **Undamped** ($c = 0$): $\beta = 0$, $\omega_1 = \omega_0$ —recovers SHM (Section 1).
- **Quasi-period** $T_1 = 2\pi/\omega_1 > T_0$ —damping always *lengthens* the period.
- **Logarithmic decrement** $\delta = \ln[x(t)/x(t + T_1)] = \beta T_1$ —how damping is measured from a decay trace.

Regime	Condition	General solution	Behavior
Overdamped	$c^2 > 4mk$	$c_1 e^{r_1 t} + c_2 e^{r_2 t}, \quad r_1, r_2 < 0$	monotone decay
Critically damped	$c^2 = 4mk$	$(c_1 + c_2 t) e^{-\beta t}$	fastest non-oscillatory
Underdamped	$c^2 < 4mk$	$e^{-\beta t} (c_1 \cos \omega_1 t + c_2 \sin \omega_1 t)$	decaying oscillation
Undamped	$c = 0$	$c_1 \cos \omega_0 t + c_2 \sin \omega_0 t$	pure oscillation

Step-by-Step Methodology

1. Write $m\ddot{x} + c\dot{x} + kx = 0$.
2. Form the characteristic equation $mr^2 + cr + k = 0$.
3. Compute $D = c^2 - 4mk$ (or compare c to $c_{\text{cr}} = 2\sqrt{mk}$).
4. Classify: $D > 0$ overdamped, $D = 0$ critical, $D < 0$ underdamped.
5. Write the matching general solution from the regime table.
6. Apply the initial conditions for c_1, c_2 .
7. (Underdamped) repackage as $A e^{-\beta t} \cos(\omega_1 t - \phi)$; report ω_1 , quasi-period, envelope.

Worked Example A — Underdamped IVP

Solve $\ddot{x} + 2\dot{x} + 5x = 0$ with $x(0) = 2$, $\dot{x}(0) = 2$.

Step 1 — Characteristic equation :

$$r^2 + 2r + 5 = 0 \Rightarrow r = \frac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i.$$

Step 2 — Read off the parameters (complex roots $\Rightarrow$ underdamped):

$$\beta = 1, \quad \omega_1 = 2, \quad x(t) = e^{-t}(c_1 \cos 2t + c_2 \sin 2t).$$

Step 3 — Apply $x(0) = 2$:

$$x(0) = c_1 = 2.$$

Step 4 — Differentiate and apply $\dot{x}(0) = 2$:

$$\dot{x}(t) = e^{-t}[(-c_1 + 2c_2) \cos 2t + (-2c_1 - c_2) \sin 2t], \quad \dot{x}(0) = -c_1 + 2c_2 = 2 \Rightarrow c_2 = 2.$$

Step 5 — Particular solution and amplitude-phase :

$$x(t) = e^{-t}(2 \cos 2t + 2 \sin 2t) = 2\sqrt{2} e^{-t} \cos\left(2t - \frac{\pi}{4}\right).$$

Envelope $\pm 2\sqrt{2} e^{-t}$, **quasi-period** $T_1 = \pi$; **the oscillation dies within a few seconds.**

Worked Example B — Overdamped vs. Critically Damped

Both start from $x(0) = 1$, $\dot{x}(0) = 1$; only the damping differs.

Overdamped: $\ddot{x} + 5\dot{x} + 6x = 0$.

Step 1 — Roots :

$$r^2 + 5r + 6 = (r + 2)(r + 3) = 0 \Rightarrow r = -2, -3 \quad (D = 25 - 24 > 0).$$

Step 2 — Apply ICs to $x = c_1 e^{-2t} + c_2 e^{-3t}$:

$$c_1 + c_2 = 1, \quad -2c_1 - 3c_2 = 1 \Rightarrow c_1 = 4, \quad c_2 = -3.$$

Result :

$$x(t) = 4e^{-2t} - 3e^{-3t}.$$

Critically damped: $\ddot{x} + 4\dot{x} + 4x = 0$.

Step 1 — Repeated root :

$$r^2 + 4r + 4 = (r + 2)^2 = 0 \Rightarrow r = -2 \text{ (double)}, \quad (D = 0).$$

Step 2 — Apply ICs to $x = (c_1 + c_2 t)e^{-2t}$:

$$c_1 = 1, \quad \dot{x}(0) = c_2 - 2c_1 = 1 \Rightarrow c_2 = 3.$$

Result :

$$x(t) = (1 + 3t)e^{-2t}.$$

Neither oscillates; the critically damped mass returns to equilibrium fastest.

3. Forced Oscillations & Non-Homogeneous Systems

The Core Logic

Apply an external driving force $F(t) = F_0 \cos \omega t$. The equation becomes **non-homogeneous**:

$$m \ddot{x} + c \dot{x} + kx = F_0 \cos \omega t.$$

By linearity the full solution splits into two pieces:

$$x(t) = \underbrace{x_c(t)}_{\text{complementary (transient)}} + \underbrace{x_p(t)}_{\text{particular (steady-state)}}.$$

x_c is exactly the damped free solution of Section 2; whenever $c > 0$ it decays to zero — the **transient**. x_p persists — the **steady-state** response that the system settles into. Using **undetermined coefficients** with trial $x_p = A \cos \omega t + B \sin \omega t$ (valid when ω is *not* a root of the characteristic equation), substitution and coefficient matching give the steady-state amplitude and phase

$$x_p = C \cos(\omega t - \delta), \quad C = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}, \quad \tan \delta = \frac{c\omega}{k - m\omega^2}.$$

All Possible Cases / Variations

- **Damped forcing** ($c > 0$): x_c is transient, so the long-run motion is purely the steady-state x_p .
- **Undamped forcing** ($c = 0$): x_c never decays — superposition of two frequencies produces **beats** or **resonance** (Section 4).
- **Non-resonant** ($\omega \neq \omega_0$): standard trial $A \cos \omega t + B \sin \omega t$.
- **Duplication / resonance** ($\omega = \omega_0$, undamped): the trial collides with x_c — multiply by t (Section 4).
- **Other forcing families**: match the trial to $F(t)$ — see the table below.
- **Frequency response** $C(\omega)$ peaks near $\omega \approx \omega_0$ — the onset of practical resonance.

Forcing $F(t)$	Trial particular solution x_p
$F_0 \cos \omega t$ or $F_0 \sin \omega t$	$A \cos \omega t + B \sin \omega t$
polynomial of degree n	$a_n t^n + \cdots + a_1 t + a_0$
$F_0 e^{\alpha t}$	$A e^{\alpha t}$
$F_0 e^{\alpha t} \cos \omega t$	$e^{\alpha t} (A \cos \omega t + B \sin \omega t)$
any trial duplicating x_c	multiply by the lowest power of t that removes the overlap

Step-by-Step Methodology

1. Write $m\ddot{x} + c\dot{x} + kx = F(t)$.
2. Solve the homogeneous equation for x_c (Section 2 classification).
3. Choose the trial x_p matching $F(t)$; check for duplication with x_c and multiply by t if needed.
4. Compute $\dot{x}_p$, $\ddot{x}_p$ and substitute.
5. Match coefficients and solve the linear system for the unknowns.
6. Form $x = x_c + x_p$.
7. Apply the initial conditions to the *full* x (not x_c alone) to find c_1, c_2 .
8. Identify transient (x_c) vs. steady-state (x_p); convert x_p to amplitude-phase.

Worked Example — Damped Forced IVP

Solve $\ddot{x} + 2\dot{x} + 2x = 2\cos t$ with $x(0) = 0$, $\dot{x}(0) = 0$.

Step 1 — Complementary solution ($r^2 + 2r + 2 = 0 \Rightarrow r = -1 \pm i$, underdamped):

$$x_c(t) = e^{-t}(c_1 \cos t + c_2 \sin t).$$

Step 2 — Trial ($\omega = 1$ is not a root, so no duplication):

$$x_p = A \cos t + B \sin t, \quad \dot{x}_p = -A \sin t + B \cos t, \quad \ddot{x}_p = -A \cos t - B \sin t.$$

Step 3 — Substitute and match coefficients:

$$\begin{aligned} \cos t: & (-A) + 2B + 2A = A + 2B = 2, \\ \sin t: & (-B) - 2A + 2B = -2A + B = 0. \end{aligned}$$

Step 4 — Solve the system:

$$B = 2A \Rightarrow A + 4A = 2 \Rightarrow A = \frac{2}{5}, \quad B = \frac{4}{5}.$$

Step 5 — Steady-state amplitude—phase:

$$x_p = \frac{2}{5} \cos t + \frac{4}{5} \sin t = \frac{2}{\sqrt{5}} \cos(t - \delta), \quad \tan \delta = 2 \Rightarrow \delta \approx 1.107 \text{ rad}.$$

Step 6 — Apply ICs to $x = x_c + x_p$:

$$\begin{aligned} x(0) = c_1 + \frac{2}{5} = 0 & \Rightarrow c_1 = -\frac{2}{5}, \\ \dot{x}(0) = (-c_1 + c_2) + B = 0 & \Rightarrow c_2 = -\frac{6}{5}. \end{aligned}$$

Result:

$$x(t) = \underbrace{e^{-t} \left(-\frac{2}{5} \cos t - \frac{6}{5} \sin t \right)}_{\text{transient} \rightarrow 0} + \underbrace{\left(\frac{2}{5} \cos t + \frac{4}{5} \sin t \right)}_{\text{steady-state}}.$$

After a few seconds only the steady-state survives, oscillating at the driving frequency with amplitude $2/\sqrt{5} \approx 0.894$.

4. Resonance and Beats**The Core Logic**

Strip the damping ($c = 0$) and drive the system: $m\ddot{x} + kx = F_0 \cos \omega t$. Now x_c never decays, so the relationship between the driving frequency ω and the natural frequency ω_0 dominates the long-run behavior.

Pure resonance ($\omega = \omega_0$). The trial $\cos \omega_0 t$ already solves the homogeneous equation, so it must be multiplied by t . The result grows without bound:

$$x_p = \frac{F_0}{2m\omega_0} t \sin \omega_0 t \implies \text{amplitude} \pm \frac{F_0}{2m\omega_0} t \text{ (linear blow-up)}.$$

Beats ($\omega \approx \omega_0$ but $\omega \neq \omega_0$, started from rest). Two nearby frequencies superpose; a product-to-sum identity reveals a slow envelope riding a fast carrier:

$$\cos \omega t - \cos \omega_0 t = 2 \sin \left(\frac{\omega_0 - \omega}{2} t \right) \sin \left(\frac{\omega_0 + \omega}{2} t \right),$$

the familiar “wah-wah” amplitude modulation. **Practical resonance** (damped) is the bounded cousin: the steady-state amplitude $C(\omega)$ peaks at $\omega_r = \sqrt{\omega_0^2 - 2\beta^2}$ with a finite maximum $\approx F_0/(c\omega_0)$ —sharper for smaller c .

All Possible Cases / Variations

- **Pure resonance** ($c = 0$, $\omega = \omega_0$): linear growth $t \sin \omega_0 t$, the “resonance catastrophe.”
- **Beats** ($c = 0$, $\omega \approx \omega_0$, from rest): amplitude-modulated; beat frequency $|\omega_0 - \omega|$, envelope frequency $|\omega_0 + \omega|/2$.
- **Practical resonance** ($c > 0$): bounded peak at $\omega_r = \sqrt{\omega_0^2 - 2\beta^2}$; height grows as damping shrinks.
- **Quality factor** $Q = \frac{m\omega_0}{c} = \frac{\omega_0}{2\beta}$ —measures how sharp and tall the resonance peak is.
- **Phase at resonance**: the response lags the driving force by exactly $\pi/2$.

Step-by-Step Methodology

1. Identify undamped ($c = 0$) vs. damped ($c > 0$).
2. Compare ω to ω_0 .
3. Undamped, $\omega = \omega_0$: use the resonance trial $t(A \cos \omega_0 t + B \sin \omega_0 t)$; expect linear growth.
4. Undamped, $\omega \approx \omega_0$ from rest: solve normally, then apply product-to-sum to expose beats.
5. Damped: response is bounded—locate $\omega_r = \sqrt{\omega_0^2 - 2\beta^2}$ and the peak amplitude.
6. Describe the envelope or growth behavior explicitly.

Worked Example A — Pure Resonance

Solve $\ddot{x} + 4x = 8 \cos 2t$ with $x(0) = 0$, $\dot{x}(0) = 0$.

Step 1 — Detect resonance ($\omega_0 = 2$ equals the driving $\omega = 2$):

$$x_c = c_1 \cos 2t + c_2 \sin 2t, \quad \text{trial } x_p = t(A \cos 2t + B \sin 2t).$$

Step 2 — Substitute (the t -terms cancel, leaving):

$$\ddot{x}_p + 4x_p = -4A \sin 2t + 4B \cos 2t = 8 \cos 2t \Rightarrow A = 0, B = 2.$$

Step 3 — Particular solution:

$$x_p = 2t \sin 2t.$$

Step 4 — Apply ICs to $x = x_c + x_p$:

$$x(0) = c_1 = 0, \quad \dot{x}(0) = 2c_2 = 0 \Rightarrow c_2 = 0.$$

Result:

$$x(t) = 2t \sin 2t.$$

Amplitude envelope $\pm 2t$ grows without bound—a real spring would eventually fail.

Worked Example B — Beats

Solve $\ddot{x} + 25x = 9 \cos 4t$ with $x(0) = 0$, $\dot{x}(0) = 0$.

Step 1 — Non-resonant trial ($\omega_0 = 5 \neq \omega = 4$):

$$x_p = A \cos 4t, \quad \ddot{x}_p = -16A \cos 4t \Rightarrow -16A + 25A = 9A = 9 \Rightarrow A = 1.$$

Step 2 — General solution :

$$x = c_1 \cos 5t + c_2 \sin 5t + \cos 4t.$$

Step 3 — Apply ICs from rest :

$$x(0) = c_1 + 1 = 0 \Rightarrow c_1 = -1, \quad \dot{x}(0) = 5c_2 = 0 \Rightarrow c_2 = 0.$$

Step 4 — Combine and apply product-to-sum :

$$x(t) = \cos 4t - \cos 5t = 2 \sin\left(\frac{9}{2}t\right) \sin\left(\frac{1}{2}t\right).$$

Fast carrier $\sin(4.5t)$ **inside a slow envelope** $\pm 2 \sin(0.5t)$ — **beats recur with period** 2π .

5. Quick-Reference Decision Map

Given $m\ddot{x} + c\dot{x} + kx = F(t)$, ask in order:

1. **Is there forcing** $F(t)$? *No* $\rightarrow$ free vibration (go to step 2). *Yes* $\rightarrow x = x_c + x_p$; find x_c via steps 2–3, then build x_p by undetermined coefficients (step 4).
2. **Is** $c = 0$? *Yes* $\rightarrow$ **SHM**: $x = A \cos(\omega_0 t - \phi)$ with $\omega_0 = \sqrt{k/m}$.
3. $c > 0$: compute $D = c^2 - 4mk$ (or compare c to $c_{cr} = 2\sqrt{mk}$):
 - $D < 0$ **underdamped** $\rightarrow e^{-\beta t}(c_1 \cos \omega_1 t + c_2 \sin \omega_1 t)$ (decaying oscillation).
 - $D = 0$ **critically damped** $\rightarrow (c_1 + c_2 t)e^{-\beta t}$ (fastest, no overshoot).
 - $D > 0$ **overdamped** $\rightarrow c_1 e^{r_1 t} + c_2 e^{r_2 t}$ (monotone decay).
4. **Forced** ($F \neq 0$): pick the trial matching F . If it duplicates x_c , multiply by t . Apply ICs to the *full* x .
5. **Driving near the natural frequency?**
 - Undamped & $\omega = \omega_0 \rightarrow$ **pure resonance**: $\frac{F_0}{2m\omega_0} t \sin \omega_0 t$ (unbounded).
 - Undamped & $\omega \approx \omega_0$ from rest $\rightarrow$ **beats** (amplitude modulation).
 - Damped $\rightarrow$ bounded steady state; **practical resonance** peak at $\omega_r = \sqrt{\omega_0^2 - 2\beta^2}$.

Diagnostic	Condition	Solution path / behavior
Free, undamped	$F = 0, c = 0$	SHM, $A \cos(\omega_0 t - \phi)$; oscillates forever
Free, overdamped	$F = 0, c^2 > 4mk$	two real $r < 0$; monotone decay
Free, critical	$F = 0, c^2 = 4mk$	$(c_1 + c_2 t)e^{-\beta t}$; fastest return
Free, underdamped	$F = 0, c^2 < 4mk$	$e^{-\beta t} \cos(\omega_1 t - \phi)$; decaying oscillation
Forced, damped	$F \neq 0, c > 0$	transient $x_c \rightarrow 0$ + steady state x_p
Forced, undamped, $\omega \neq \omega_0$	$F \neq 0, c = 0$	superposition; beats if $\omega \approx \omega_0$
Forced, undamped, $\omega = \omega_0$	$F \neq 0, c = 0$	pure resonance; amplitude $\propto t$

Many systems sit between regimes—a lightly damped driven oscillator shows a sharp but *finite* resonance peak, the smooth bridge between pure resonance and an ordinary forced response. Always identify ω_0 , β , and ω first; those three numbers decide everything.